

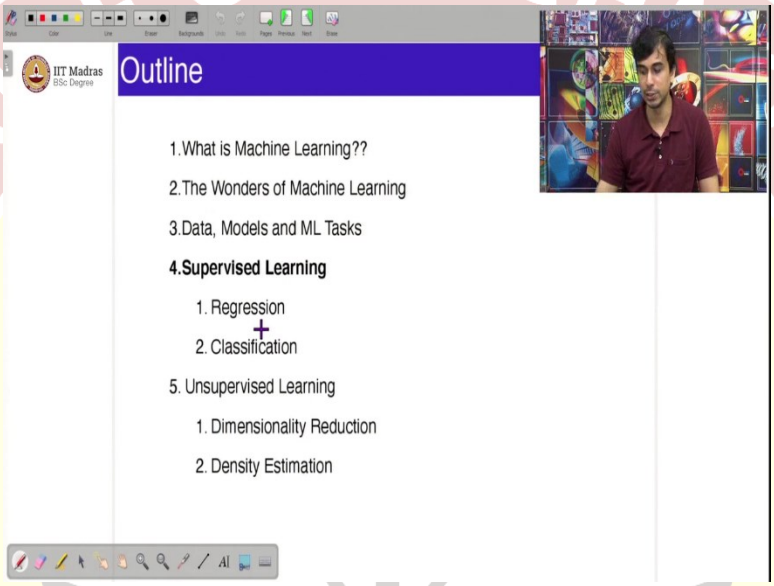
IIT Madras

ONLINE DEGREE

Machine Learning Foundations
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Supervised Learning - Regression

Hello, everyone, and welcome to another lecture on Machine Learning Foundations.

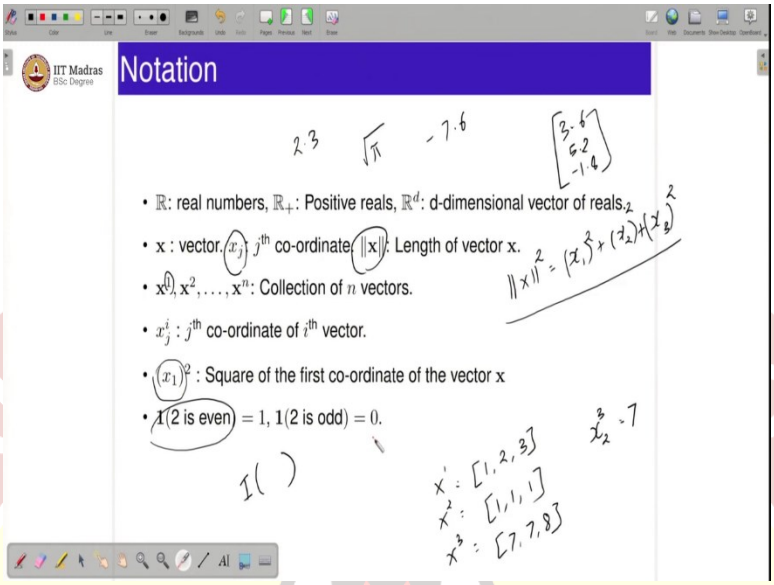
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In this lecture, we will take a deeper dive into one main paradigm of machine learning, which is supervised learning. There are several supervised learning tasks. But the most, two most important tasks that are encountered in machine learning are regression and classification. And in fact, supervised learning is the most common paradigm of machine learning.

And when someone says, machine learning, they typically mean supervised learning unless they qualify it with extra things like unsupervised learning or reinforcement learning and so on. In this lecture, we will take a deeper dive into regression and classification, which are types of supervised learning problems.

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So, before we do that, we will set up some notation, which we will need to handle the rest of the lecture. We will use $\mathbb{R}$ to denote the set of real numbers. Pretty much anything that you can, any scalar is a real number. So, for example, you have 2.3 is a real number, $\sqrt{\pi}$ is a real number, - 7.6 is a real number. All of these are examples of real numbers. The positive reals are simply real numbers that are positive.

And another set that you will encounter often is $\mathbb{R}^d$, which is the set of d dimensional vector of reals. For example, an element in $\mathbb{R}^3$ might be something like $\begin{bmatrix} 3.6 \\ 5.2 \\ -1.8 \end{bmatrix}$ is an example of an element in $\mathbb{R}^3$. We will often denote a vector by an element like x and x subscript j will denote the jth coordinate of the vector and norm of the vector or the length of the vector is denoted by this particular symbol, which the length will always mean the euclidean length.

For example, if we have norm of a three dimensional vector x is given by, the norm square is easier to say, the norm square is simply $x_1^2 + x_2^2 + x_3^2$, norm of this, norm or the length of this vector x is simply the $||x|| = \sqrt{x_1^2 + x_2^2 + x_3^2}$.

In a lot of situations, we will be working with a collection of vectors and we will denote different elements of the collection by a superscript. So, note that we are using superscript for denoting different vectors and subscripts for denoting elements in the vector. So, we can put both of these

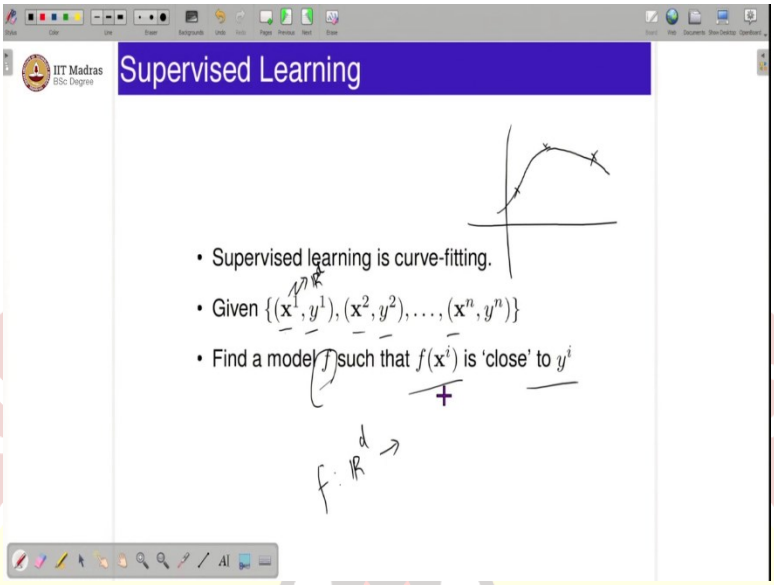
together to have, for example, something like x_j^i. x_j^i corresponds to the jth coordinate of the ith vector. So, for example, let us say you have a collection of three vectors in let us say 3D space. So, let us say you have $x^1 = [1,2,3], x^2 = [1,1,1], x^3 = [7,7,8]$.

You can have x_2^3 is the second element of the third vector which would be 7. Note that this is different from the $(x_2)^3$. We will denote that differently by, we will make a distinction between that by trying to use parenthesis around x1. So, $(x_1)^2$ corresponds to the square of the first coordinate of the vector x. For example, here, you can see that to denote norm x square, I purposefully created $(x_1)^2$. I might forget to do that some×, but from the context, it should be clear whether I am talking about the square of the x1 or the second vector.

From context it should be clear whether we are talking about superscripts or powers. This confusion is inevitable, because we just do not have enough space for denoting everything that we will have. But it should be clear from context whether we are talking about superscripts or powers. And one other notation that we will be using often is indicator variables, which is denoted by 1 of something. So, you can have a predicate class which is either true or false and based on that the output of this is going to be either 1 or 0. For example, you can have 2 is even.

2 is even is a class which is known to be true. So, $1(2\ is\ even) = 1$, but indicator of 2 is odd is known to be false. So, $1(\ 2\ is\ odd) = 0$. So, this is another notation which will be very useful for converting from math, from English to math in a lot of places. So, this is the most basic condition that you will need for handling the rest of the lecture. We will use other notation which we will introduce as and when required.

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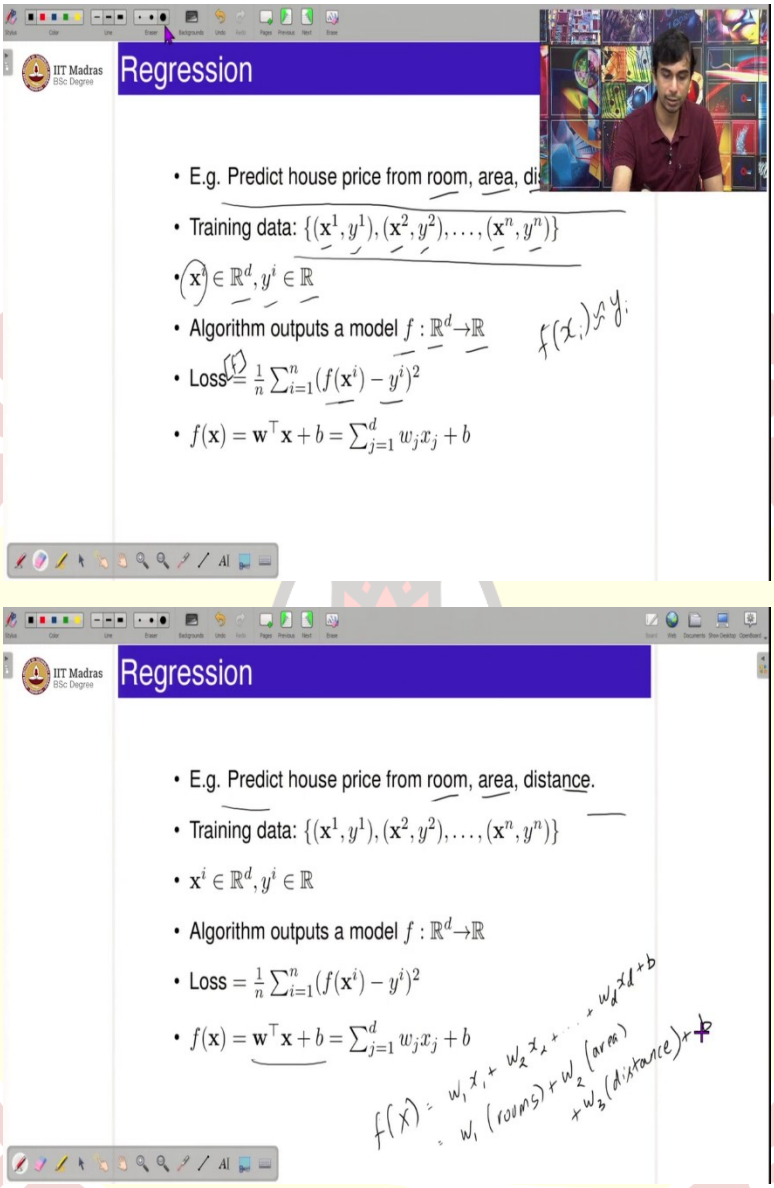


So, now, we enter the supervised learning paradigm. What is supervised learning? You can simplify, you can extreme, there is an extreme simplification that you can do and say that supervised learning is simply curve-fitting. Curve-fitting is something which you all would have done in your school. For example, you might have done something like this. You might have points like this, and you might fit a curve through these points. This is an example of a curve that fit, you fit a curve through these points.

Supervised learning at its core, at its essential is just curve-fitting. There are several aspects to this which make the problem much more complicated. But at its core, it is just curve-fitting. Which means that you have, what you have is a bunch of points, which are $\{(x^1, y^1), (x^2, y^2), (x^3, y^3), \ldots (x^n, y^n)\}$. Each of these x^i are a vector, let us say in $\mathbb{R}^d$. It is a d dimensional vector. And y^is are in some space. Typically, they will be either reals or + 1 or - 1 or some small set. We will come to that later.

The goal is to find a model. A model in this case simply is a function f from $\mathbb{R}^d$ to the space of labels. So, each data point x^i, y^i has an instance x^i and a label y^i. Your goal is to find a model f such that f of xi is close to y^i. And this is what a function really represents the curve here and you want the code to be as close as possible to the points. And that is represented by saying that you want to find the model f such that $f(x^i)$ is close to y^i. This is being intentionally vague because you will see this instantiated for regression and classification soon.

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For the case of, what is an example regression problem, we saw that we want to predict let us say house price from room area and distance. So, that is an example of a place where a regression model is used. And how would you build such a model? Well, you will need training data, you will need a data set, which will consists of house one and the price of the house one, house two and price of house two, house three and price of house three.

When we say house one, we mean simply a three dimensional vector giving its number of rooms, its area and the distance to metro. So, each house is represented by a three dimensional vector and

you have a label, which corresponds to the price of the house. So, this would be an example of training data.

You have your instances x^i in the general setting, it would be in a d dimensional space and your labels y^i would be real valued. Your learning algorithm would essentially output a model f, which is essentially a function from $\mathbb{R}^d$ to $\mathbb{R}$. And how would you evaluate this model. So, ideally, you want your model to be such that $f(x^i)$to be approximately equal to y^i. So, if $f(x^i) = y^i$ for all i, then that is the best model that you can think of right now at least.

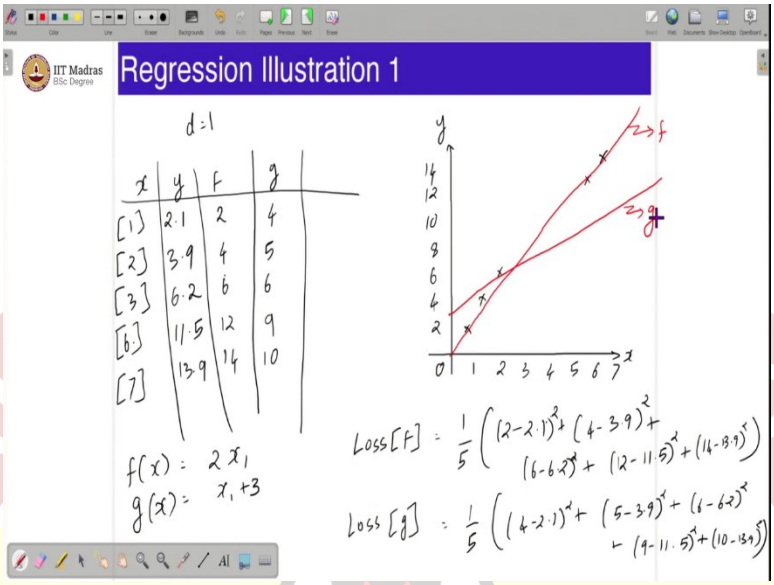
So, but that might not always be possible. You might be forced to choose a model which is not perfect. And a typical loss that is used for evaluating this model, evaluating a given model f, let us call that loss of f is simply measures the deviation of f of x^i from y^i by a simple square thing. So, if $(f(x^i) - y^i)$the whole square essentially captures how far is f of xi from y^i. You do not want your f of x^i to be too much above y^i or it should not be too much below y^i also. The ideal situation is f of xi is exactly equal to y^i, but you are willing to tolerate a little bit of error.

And $f(x^i) - y^i$ the whole square essentially ensures that it is always positive. This loss can never be negative. And immediately, you can see that when loss of f is equal to 0, then f of xi is equal to yi for all i. That is the best, that is the smallest loss possible. Typically, what you were, you like your learning algorithm to do is to find a model f which has low loss. So, that is the goal of a learning algorithm.

I have been saying f as a function from Rd to R, but how do you represent such a function? The most common representation for a function for such a function is a linear parameterization. That is, you are going to say my function f is given by $w^T x + b$. What do I mean by that? Well, f of x is going to be equal to $w^1 \times x^1 + w^2 \times x^2 + \ldots, w^d \times x^d + b$. In this case, your w^1, w^2. These are your parameters. w^1, w^2, w^d and b are your parameters and these, the dependence of the price or the output model is linear in terms of its input, actually, it is called a linear model.

For example, in the case of the house prediction price, it might say something like this. It is $w^1 \times$number of rooms + $w^2 \times$area + $w^3 \times$the distance + b. This is an example of a linear model. The w^1, w^2, w^3 and b form the parameters. This represents a linear function. So, every possible value of w^1, w^2, w^3 and b corresponds to a model. What your learning algorithm does is essentially find the best possible w^1, w^2, w^3, b for this particular training data.

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So, here is a simple illustration of how that might be done. So, let us have a very simple data where your input dimension d is simply just one dimension, let us say d is equal to 1. Let us have a very simple data here, let us say x, y. So, your x is just a one dimensional vector. So, the instances are one dimensional vector.

So, effectively they are scalars. But I am going to write them as a vector format so that you understand that x can be a vector. Let us say the first data point x is x1 is, let us say just 1. The second data point, let us say it is equal to 2, let us say the third data point is 3, the fourth data point is say 6. The fifth data point is 7.

Now, five data points, x_1, x_2, x_3, x_4, x_5, they are all d dimensional vectors, but d happens to be equal to 1. So, this is a very simple case. And your y let us say it is 2.1 for y_1 is let us say 2.1, say 3.9, y_3 is let us say 6.2, y_4 is let us say 11.5 and let us say y_5 is 13.9. In this particular case, you can actually easily visualize the data because x happens to be just one dimensional. So, you can just plot the data and see what we mean by that. So, x which is 0, 1, 2, 3, 4, 5, 6, 7, let us say 2, 4, 6, 8, 10, 12, 14.

So, you have the first data point is 1,2.1 that approximately would be here, 2,3.9 would be approximately here, 3,6.2 would be approximately here, 6,11.5 would approximately be here, and 7,13.9 would approximately be here. So, now you have your, this is your x-axis, this is your y-

axis. Because the data happens to be just d equal to one dimensional, you can do the simple plotting to visualize the data.

What would a learning algorithm do? Learning algorithm would take in this data and output a model f which has small loss. We will not actually do the full business here. But we will just try to evaluate two distinct models. Let us have two distinct models. Let us evaluate the model given by f. Let us say f of x is let us say it is just $2x_1$. So, even though x is just a scalar, I am just going to write $2x_1$ here for completeness. And let us say another model, g of x is, let us say, $x_1 + 3$.

So, now, in general, the learning algorithm would have to find the best model. I am going to make the job easier for the learning algorithm and say that you just choose the best out of these two models. How would the learning algorithm do that? It is a very simple thing where it can just compute f and compute g, and decide which of these models have a, has a smaller loss. So, for example, if you compute f, it is $2x_1$. So, this is 2, this is 4, this is 6, this is 12 and this is 14. g would be x1 + 3 it would be 4, 5, 6, 9 and 10.

What is the loss of f? Well, that is simply equal to $\frac{1}{5} \times (2 - 2.1)^2 + (4 - 3.9)^2 + (6 - 6.2)^2 + (12 - 11.5)^2 + (14 - 13.9)^2$. So, this is the loss of the model f. You can compute this loss. And similarly, you can compute the loss of g also.

Or would that be it is the average of these terms which is $(4 - 2.1)^2 + (5 - 3.9)^2 + (6 - 6.2)^2 + (9 - 11.5)^2 + (10 - 13.9)^2$. You can compute the loss of the models f and g. And clearly, you can see that the loss of f is going to be much lesser than the loss of g. And your learning algorithm would prefer model f over model g.

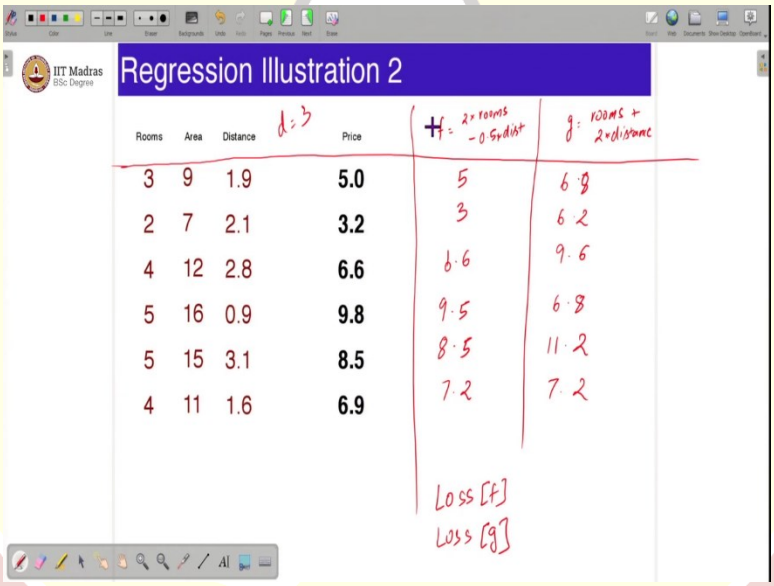
Is that the best model? We do not know yet. I mean, the learning algorithm in the real world, in practice, would not just have to choose between two models f and g. I have done here as just an illustration to show you that f, I mean, in principle, the learning algorithm just chooses the best model from an entire library instead of just the best out of two models.

And so, in this particular case, we can actually do a better job and actually illustrate the function f. And what is the function f here, the function f is simply $2x_1$ and that you can actually plot that function here. And if you do that, that is an example. This is f. It is not particularly straight, but I cannot do a much better job than this. So, this is f.

And g would essentially be $x_1 + 3$, which would be something like this. This is, you can clearly see that f is, f fits the data points much better than g. So, f is a much better curve that fits the data points than g and that is the reason why you would have loss of f is less than loss of g and the learning algorithm would prefer f over g. So, this is a very simple illustration of regression.

The learning algorithm, of course, we have, I have not given you the learning algorithm. The learning algorithm in principle does this job of finding the best model over an infinite set. So, here we have just done the, make the, made a simplified learning algorithm where it just chooses the best among two models. But in the real world, it has to choose the best among an entire library of models.

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Here is another simple illustration that to make things more concrete for the case of price illustration. So, you can have, let us say, two models. The first model f can be something like let us say, 2 × rooms - 0.5×distance. That is one possible model. You can have another model, let us say, g, which is rooms + 2×distance. This is another simple example here. Now, d is equal to 3. So, you have your data lives in three dimensional space, and of course, the labels are always going to be real. They are going to be scalar. So, that is your 5, 3.2, 6.6, and so on.

And now, let us say the learning algorithm, once again, we are making the job of the learning algorithm easy. And we are just asking the learning algorithm to choose the better among these

two models, f and g. How would we do that? Well, once again, you can repeat the same thing that is you can compute the f and g for the training points and see which has the smaller loss.

For example, if you do that for f 2×rooms - 0.5×distance, that would be 6 - about approximately, I am not going to write the exact numbers here, because it does not matter. I am just going to write an approximate number here, that is approximately 5. 2×rooms - 0.5×distance it is 2×2, 4 - 0.5×2.1 that is approximately 3.

Similarly, you have 2×rooms, which is 2×4 - 0.5×2.8, 8 - 1.4 that is approximately 6.6. Similarly, you can do that for the other three data points also. So, 10 - 0.45 that is approximately 9.5, that is 10 - 1.5 that is approximately 8.5, 8 - 0.8 that is approximately, let us say 7.2. So, this is the predictions made by model f.

And you can do the same thing for model g that would be, let us say, rooms + 2×distance, that is 3 + 3.8, let us say, 6.8, 2 + 4.2 that is 6.2, 4 + 5.6, 9.6, 5 + 1.8 that is 6.8, 5 + 6.2 that is 11.2, 4 + 3.2 that is 7.2. Now, this column gives the predictions of model f and this column gives the predictions of model g. You can use this to compute loss of f and loss of g.

How would you do that? Well, that is the same thing that you average the square difference between 5 and 5.0, 3 and 3.2, 6.6 and 6.6, 9.5 and 9.8, 8.5 and 8.5, 6.9 and 7.2, and so on, that would give you the loss of f. And similarly, you can do the same thing for loss of g. And you can clearly once again see that loss of f is going to be much smaller than loss of g. So, f is a much better model than g. And once again, you can see that it is saying a very meaningful thing. It is f is saying that 2×rooms - 0.5×distance is a good estimate of the price, which is a reasonable, once you see the answer it looks reasonable.

That is, well, it seems to depend on the number of rooms, it increases with the number of rooms, but it does decrease with the distance to the metro. That is what this model is saying. And it does seem to work well on this data. If the learning algorithm has just given these two models to choose from, it will choose f. But once again the learning algorithm is not given just these two models, it is given an entire library of models and it will choose the best out of this library of models. So, with that, I think we have wrapped up regression. We will move on to classification in our next video.